



Cambrian College

School of Business and Information Technology, Creative Arts, Design, Music and Hospitality

Course Outline

Cambrian College would like to acknowledge that we are situated on the traditional lands of the Robinson Huron Treaty, and acknowledge our host, Atikameksheng Anishnawbek, as well as our ancestors of this land, and the Anishnaabe people.

Course Title	Enterprise Analytics				
Course Code:	ANA1005	Credit Value:	3	Credit Hours:	42
Programs:	BAPG Business Analytics				
Equivalencies		Prerequisites		Corequisites	

This course may be delivered in a variety of different formats: 100% in-class, 100% online (or a blend of both), videoconferencing, distributed learning or off-campus. Please confirm with your faculty member which format will be used for your section of this course.

General Education Course: ☐

Degree Breadth Course: ☐

Eligible for PLAR: ☒

COURSE DESCRIPTION

In this course, students will discover how to identify the data resources available to an organization and evaluate the quality of these data sources. Students will explore the processes for validating data, cleansing data, and ensuring completeness of data using a variety of methods.

Date: September 07, 2021

Prepared By: Sidney Shapiro

Approved by:

JILL FERGUSON
Chair, School Of Business And Information Technology,
Creative Arts, Design, Music And Hospitality

Effective: Fall 2021, Winter 2022, Spring 2022

RELATIONSHIP TO PROGRAM VOCATIONAL LEARNING OUTCOMES

PROGRAM LEVEL	
This course contributes to your program by allowing you to demonstrate the following vocational learning outcomes:	
Program(s)	Vocational Learning Outcomes
Business Analytics	College Standards 1. Collect, manipulate and mine data sets to meet organizational need. 2. Recommend different systems architectures and data storage technologies to support data analytics. 3. Design and apply data models that meet the needs of a specific operational/ business process. 4. Develop software applications to manipulate data sets, correlate information and produce reports. 5. Apply data analytics, business intelligence tools and research to support evidence-based decision making. 6. Identify and assess data analytics business strategies and workflows to respond to new opportunities or provide project solutions.

COURSE CURRICULUM**Topics/Concepts Covered in This Course**

- Importing data
- Cleaning and validating data
- Testing for Normality
- Identifying missing data
- Transforming data

COURSE LEVEL: Learning Outcomes and Objectives	
To earn credit for this course, you must reliably demonstrate your ability to:	
Learning Outcome	Objectives
1. Import data from various sources	1.1 Identify appropriate sources of data 1.2 Acquire, import, and export data in various formats 1.3 Adapt data from various sources
2. Clean and validate data	2.1 Identify optimal data cleaning strategies 2.2 Clean, filter, and edit data based on requirements 2.3 Analyze data in Python, R, and Excel 2.4 Working with text, dates, and strings 2.5 Ensure data quality and validity

Learning Outcome	Objectives
3. Test for Normality	3.1 Explain why normally distributed data is important for many statistical analyses. 3.2 Calculate the skew of a dataset and interpret the values. 3.3 Discuss how ceiling and floor effects can affect data analysis. 3.4 Construct and interpret histograms and other plots to assess normality
4. Identifying missing data	4.1 Ensure data completeness 4.2 Perform mean and ratio imputation 4.3 Impute missing data using linear regression 4.4 Compare and contrast the pros and cons of different imputation methods. 4.5 Identify where it is appropriate to use each imputation method.
5. Transform data	5.1 Transform skewed data 5.2 Explain the ETL (extract, transform and load) Process 5.3 Discuss challenges associated transforming big data

Essential Employability Skills

Communication

- communicate clearly in written, spoken, and visual form that fulfills purpose/needs of audience.
- respond to written, spoken, or visual messages in a manner that ensures effective communication.

Numeracy

- execute mathematical operations accurately.

Critical Thinking and Problem Solving

- apply a systematic approach to solve problems.
- use a variety of thinking skills to anticipate and solve problems.

Information Management

- locate, select, organize, and document information using appropriate technology and info systems.
- analyze, evaluate, and apply relevant information from a variety of sources.

Interpersonal

- show respect for the diverse opinions, values, belief systems, and contributions of others.

Personal

- manage the use of time and other resources to complete projects.
- take responsibility for one's own actions, decisions, and consequences.

Delivery Method

- Classroom: Course is delivered through scheduled synchronous teaching that may be face-to-face and/or virtual.
- Online: Course is fully delivered through asynchronous teaching.
- Hybrid: Course combines scheduled synchronous and unscheduled asynchronous teaching.
- HyFlex: Course includes both synchronous and asynchronous learning and the student can move between both components seamlessly.

Learning Activities

- Lectures
- Class Discussions
- Labs
- Research
- Self-Directed Learning
- Case Studies
- eLearning Components

Evaluation Plan

Grading Scheme

A	80% - 100%
B	70% - 79%
C	60% - 69%
D	50% - 59%
F	0% - 49%

Evaluation Method	Value (%)
Applied Activities (G)	15%
Case study. Students will apply their knowledge to clean, analyze, and report on a given data set. Please see instructor for details.	
Test/Exam	20%
There will be multiple exams in this course. Please see your instructor for details.	
Assignments	45%
There will be multiple assignments in this course. Please see your instructor for details as the assignments may be in the form of written case studies, presentations, individual or group projects, or another form not listed.	
Quizzes	20%
There will be multiple quizzes in this course. Please see your instructor for details.	

ADDITIONAL INFORMATION

College

Modifications to the Course Evaluation Plan

Under exceptional circumstances, Cambrian reserves the right to alter the Course Evaluation Plan. Upon approval from the Dean, faculty members will notify students of any modification to the Course Evaluation Plan and post to the course Learning Management System site (Moodle).

Academic Policies

Students must adhere to all academic policies. These are available on myCambrian as well as the College website at <https://cambriancollege.ca/about/official-documents-and-policies/academic-policies/>.

Accessibility Services

Cambrian is committed to creating an inclusive learning environment where we support our students' success. Students who are registered with the Glen Crombie Centre can access accessibility and/or counselling services by:

1. Registering and scheduling an appointment through Clockwork in the Student Portal in myCambrian.
2. Contacting 705-566-8101 ext. 7420 or 7311, or
3. E-mailing accessibilityservices@cambriancollege.ca or Counselling@cambriancollege.ca

Attendance

Regular attendance is strongly encouraged as it contributes to student success. Some courses have specific lab/participation requirements. Students are expected to be aware of, and adhere, to these requirements.

Audio/Visual Capture

Sounds and images from this class, and contributions made by a participant, virtually or in-person, are recorded under the authority of the Ontario Colleges of Applied Arts and Technology Act, 2002. The main purpose for these recordings is to allow students enrolled in this course, whether they attend the particular class in person, virtually, or at all, to review class content and activity. The use of class recordings is for personal use only and shall not be shared or transferred.

These recordings may also be reviewed by faculty members to prepare future classes, to evaluate students, to collaborate in program/course development/review, or to provide feedback to students or other faculty. Any questions about this collection may be addressed to your respective Dean.

Copyright

Students may not submit work or partake in activities that infringe on the Canadian Copyright Act, or the related Cambrian College Fair Dealing Guidelines and Copying Guidelines. The guidelines may be viewed at: <http://cambriancollege.libguides.com/copyrightinfo> or by contacting the Library for questions related to copyright: library@cambriancollege.ca

Wabnode Centre for Indigenous Services

Students are encouraged to come to Wabnode to meet our Elders and to access personal counselling and connect with academic tutoring. Our team is also able to provide students with information related to funding and scholarship and bursary opportunities for Indigenous students. We encourage students to come to Wabnode and find out about opportunities to share in cultural, sports and community activities. Students can also contact Wabnode at wabnode@cambriancollege.ca

Out-of-Class Assistance

The faculty will inform students of their Out-of-Class availability through the Learning Management System site (Moodle).

Prior Learning Assessment and Recognition (PLAR)

Students wishing to have work or life experience considered through Prior Learning Assessment and Recognition should contact pathways@cambriancollege.ca

Testing Practices at Cambrian

Many courses include major tests and/or final exams. The practice at Cambrian requires that these types of test situations involve proctoring to ensure academic integrity. Online tests/exams may employ a proctoring service to enable you to take your exam from a location of your choosing within a period specified by your instructor. When you are taking an online test/exam, the proctoring service may capture your video, screen, audio and web surfing particulars to protect academic integrity. Cambrian College collects, uses, discloses and retains personal information in compliance with the Freedom of Information and Protection of Privacy Act (FIPPA). Your personal information is being collected under the authority of the Ontario Colleges of Applied Arts and Technology Act S.O. 2002, c.8, Sched. F. This information will be used for the purpose of administering a test/exam through an online proctoring service acting as an authorized agent of the College. Please refer to Cambrian's Confidentiality of Student Records Policy for more details. If you have any questions regarding the collection of your personal information, please contact Vice President Academic, Cambrian College, 1400 Barrydowne Rd., Sudbury ON P3A 3V8, 1-705-566-8101.

Tests and Evaluations

Students are required to write tests and/or complete evaluations as scheduled. Exceptions may be made in the event of an emergency or a sanctioned event (e.g., varsity sports, field trip, religious observances, etc.). In non-emergency situations, students are expected to contact the faculty member in the event that they can not be present for a test/evaluation to explain the reason for their absence. At the discretion of the faculty member, an alternate time for the test/evaluation may be allowed.

Transfer Credit

Students wishing to have courses from other programs or institutions assessed for equivalency and/or transfer credit should contact transfercredits@cambriancollege.ca.

Equity, Diversity and Inclusivity

Cambrian is committed to building and preserving an equitable, diverse, and inclusive learning community where students, faculty, and staff may achieve their full potential in an environment characterized by equality of respect and opportunity. All students and employees have the right to live and work in an environment that is free from discrimination and harassment. Therefore, Cambrian College will not tolerate any form of discrimination or harassment in its employment, education, accommodation, or business dealings. For more information, please visit: <https://cambriancollege.ca/about/official-documents-and-policies/equity-human-rights-and-accessibility/>